



Thin-Layer Chromatography and Real-Time Coupled Assays to Measure ATP Hydrolysis

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Abstract

Many chemical reactions in the cell are thermodynamically unfavorable. To overcome this barrier, the energy released from the hydrolysis of adenosine triphosphate (ATP) is coupled to these reactions via ATP hydrolyzing enzymes known as ATPases. These enzymes are ubiquitous in nature and frequently act as molecular motors in processes ranging from DNA replication to protein degradation. Assays that characterize ATPase activity in vitro are important tools to gain insight into their functions in vivo. Here, we describe a direct and flexible thin-layer chromatography method for detecting ATPase activity using radiolabeled ATP. Additionally, we describe a high-throughput coupled reaction assay pairing ATP hydrolysis with nicotinamide adenine dinucleotide (NADH) oxidation to monitor ATP hydrolysis in real time.

Key words ATPase, NADH, Thin-layer chromatography (TLC), Coupled assay

1 Introduction

ATPases are enzymes that catalyze the hydrolysis of adenosine triphosphate (ATP) into adenosine diphosphate (ADP) and inorganic phosphate [1]. This reaction releases energy that is coupled by the ATPase to drive chemical reactions or conformational changes in the protein itself or other molecules. The hydrolysis of ATP by these enzymes is a conserved biochemical activity found in all life forms and in a diverse set of reactions, including DNA replication and transcription, cellular metabolism, ion pumping, and protein assembly and trafficking [2]. Characterizing the ATPase activity of an enzyme is a critical step in determining how it functions both in vitro and in vivo. Many techniques have been described to measure the ATPase activity of various proteins, each with their own merits and limitations. Many methodologies detect the release of inorganic phosphate using radioisotopes [3] or colorimetry [4]. These methods are useful for high-throughput screening but can suffer from a poor signal-to-noise ratio and/or dynamic detection range. The direct detection of ATP hydrolysis can be

achieved using radiolabeled ATP (e.g., $\alpha^{32}\text{P}$ -ATP) and thin-layer chromatography (TLC) [5]. This technique is sensitive and simple to perform but is not amenable to high-throughput assays. An alternate technique couples multiple enzymatic reactions together, detecting ATP hydrolysis in an indirect manner [6]. This coupled assay is simple to conduct, high throughput, and boasts automated data collection. Relative to the direct assay, large quantities of enzyme are needed for this method however, which can be a limiting factor. Here, we describe both the direct TLC assay and the NADH-coupled 96-well plate assay for the measurement of ATPase activity. The assay best suited for a particular ATPase depends on protein availability, equipment access, optimal reaction temperature, and your specific ATPase. Conducting both assays may be necessary for a thorough characterization of an enzyme's ATPase activity. However, one or the other will be suitable for most situations.

The direct TLC ATPase assay utilizes a positively charged polyethyleneimine (PEI) cellulose membrane as a stationary phase, and a formic acid and lithium chloride solution as the mobile phase. The less polar ADP migrates further in the mobile phase than ATP, thus producing separation on the membrane (Fig. 1). Detection of ATP and ADP is achieved by the use of radiolabeled ATP in the assay and subsequent phosphorimaging. The coupled ATPase assay uses two additional enzymes, pyruvate kinase and lactate dehydrogenase, to regenerate ATP and couple ATP hydrolysis to NADH oxidation (Fig. 2). ATP and ADP absorb light at similar wavelengths and, thus, cannot be easily distinguished. NADH and NAD^+ , however, have distinct absorption spectra, and can easily be distinguished using spectrophotometry. Using the reactions in the coupled assay, ATP hydrolysis is equimolar to NADH oxidation, as monitored by absorbance at 340 nm. The TLC ATPase assay benefits from the small quantity of protein needed, the ability to easily assay a wide range of temperatures, and the lack of any necessary specialized equipment [7]. The NADH-coupled assay benefits from its high throughput nature, automated data collection, ease of preparation, and ATP regeneration, which maintains a constant concentration of ATP available for the ATPase [8].

Below, we describe detailed protocols for both the direct TLC assay and NADH-coupled ATPase assay. The TLC assay involves the preparation of the chromatography chamber and plate, taking time points during the assay itself, and development of the TLC plate. The NADH-coupled assay involves the preparation of the ATPase reaction, loading of a 96-well plate, and data collection via a spectrophotometer or plate reader. Both assays are capable of characterizing the ATPase activity of an enzyme, providing valuable insight into its *in vitro* activity and *in vivo* function.

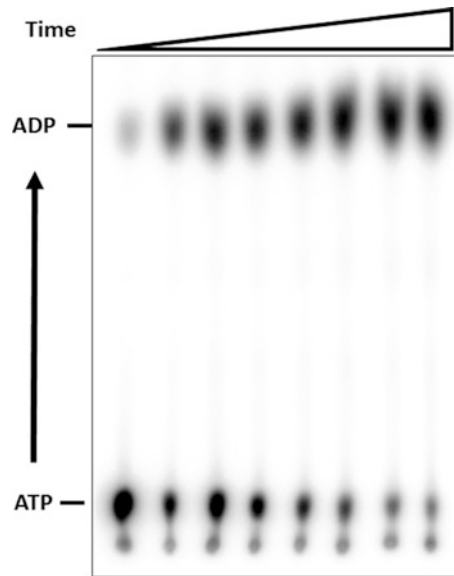


Fig. 1 Representative data of a thin layer chromatography ATPase assay. ADP migrates with the mobile phase up the TLC plate, while ATP migration is minimal. The arrow shows the direction of mobile phase movement. Over the course of the ATPase reaction, ATP is hydrolyzed into ADP and inorganic phosphate. This can be visualized by the presence of more ADP in the later time points of the assay

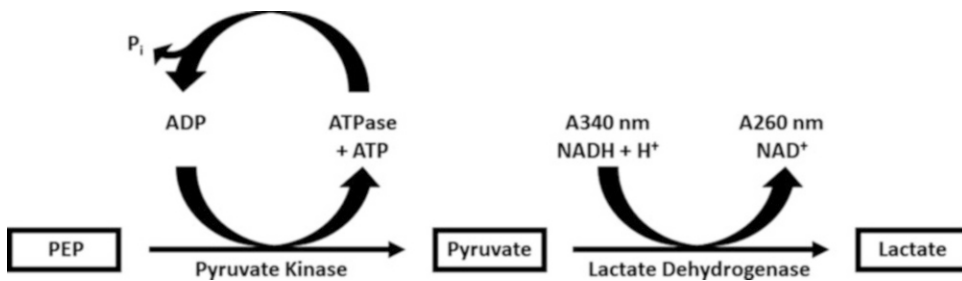


Fig. 2 The reactions taking place in the NADH-coupled ATPase assay. The ATPase converts ATP into ADP and P_i . Pyruvate kinase converts ADP back into ATP, while converting PEP into pyruvate. Pyruvate is then converted into lactate by lactate dehydrogenase. During this reaction, lactate dehydrogenase oxidizes NADH into NAD^+ , changing its absorbance peak from 340 to 260 nm. This change in absorbance can be detected by a spectrophotometer or appropriate plate reader in real time

2 Materials

All buffers and other reagents should be made with ultrapure MilliQ water (resistivity of $18.2 \text{ M}\Omega\cdot\text{cm}$ at 25°C).

2.1 TLC Assay

1. Running buffer: 1.0 M formic acid, 400 mM LiCl. Add 1.70 g of LiCl and 4.24 mL formic acid (23.6 M) to a 100 mL

graduated cylinder. Bring the volume to 100 mL with water, and store in a glass bottle (*see Note 1*).

2. 2× ATPase buffer: 50 mM Na-HEPES, pH 8, 10% glycerol, 100 mM NaOAc, pH 7.5, 300 μM NaCl, 0.02% NP-40.
3. Reaction stop buffer: 2% SDS, 20 mM EDTA.
4. α[³²P]-ATP (*see Note 2*),
5. TLC PEI cellulose sheets 20 × 20 cm.
6. Glass TLC tank (*see Note 3*).
7. Vacuum grease (*see Note 4*).

2.2 NADH-Coupled Assay

1. 2× ATPase buffer: 50 mM Na-HEPES, pH 8.0, 10% glycerol, 100 mM NaOAc, pH 7.5, 300 μM NaCl, 0.02% NP-40. Store at −20 °C.
2. 100 mM ATP, pH 7.0: 100 mM ATP in water. Adjust the pH to 7.0 with NaOH. Store at −80 °C.
3. 10 mM phosphoenolpyruvic acid (PEP): Store at −80 °C
4. 10 mM NADH, pH 8.0: Dissolve 71 mg β-Nicotinamide adenine dinucleotide in 10 mL 25 mM Na-HEPES, pH 8. Aliquot 1 mL portions into 1.7 mL tubes and store at −80 °C.
5. Rabbit pyruvate kinase.
6. 100 mM MgCl₂.
7. Lactate dehydrogenase.
8. 10 μM DNA.
9. Clear, flat-bottom 96-well plate.
10. BioTek Synergy H1 plate reader (*see Note 5*).

3 Methods

3.1 TLC Assay

1. To delineate lanes for your reactions, draw a line with a pencil across a thin-layer chromatography sheet 1 cm from the bottom edge (*see Note 6*). Starting 0.5 cm from the edge, make a mark every 1 cm along the line, making 20 marks total.
2. Add approximately 50 mL of running buffer to the TLC tank (*see Note 7*). Use vacuum grease to line the top edge of the tank (ensuring a tight seal), and cover with the lid. Let the tank equilibrate with the buffer for ~30 min.
3. Add 2 μL of reaction stop buffer to seven tubes for each reaction, on ice.
4. Add 3.75 μL 2× ATPase buffer, 3 μL of 5 μM DNA oligonucleotide, 7.5 μL of 2× concentrated protein in 1× ATPase buffer, and 3.75 μL water to a 0.7 mL tube on ice (*see Note 8*). Mix gently with a pipette, and repeat for the number of reactions

desired (*see Note 9*). Assemble an additional reaction without protein, adding 7.5 μL $1\times$ buffer instead to reach a volume of 15 μL . This will serve as a control for background ATP hydrolysis.

5. Add 0.6 μL $\alpha[^{32}\text{P}]\text{-ATP}$ (equal to 6 μCi) into each tube.
6. Keeping the reactions on ice, aliquot 2 μL of each reaction into a stop tube on ice. This is the zero time point (*see Note 10*).
7. Move the reactions to the desired temperature(s) (*see Note 11*). At 1, 2, 5, 10, 15, and 20 min, aliquot 2 μL of each reaction into a separate stop tube on ice (*see Note 12*).
8. Spot 1 μL of a stopped reaction onto a mark on the TLC plate. Repeat for each stop tube. Allow the spots to dry until no longer visible (*see Note 13*).
9. Place the dried TLC plate into the tank, submerging the bottom edge in running buffer. Ensure the tank lid is properly sealed around the edges of the tank. Develop the plate for 1 h, allowing the solvent front to reach 1 cm from the top of the TLC plate (*see Note 14*).
10. Remove the TLC plate from the tank and allow it to dry, cellulose side up (*see Note 15*).
11. Wrap the TLC plate in plastic wrap and expose to a phosphor plate for 4 h (*see Note 16*).
12. After exposure, scan the plate using a Typhoon imager. The hydrolyzed ADP will have migrated toward the top of the plate, while unhydrolyzed ATP will remain near the original spot placement (Fig. 2).
13. Quantitate the percent ATP hydrolyzed in each reaction by densitometry using the following equation: $((\text{ADP})/(\text{ADP} + \text{ATP})) \times 100$. The amount of ATP hydrolyzed in the no protein control reactions should be subtracted from each reaction.

3.2 NADH-Coupled Assay

1. Preheat the plate reader to 37 $^{\circ}\text{C}$.
2. Assemble 60 μL of the $2\times$ reaction cocktail for each reaction condition on ice: 10 mM ATP, pH 7.0, 10 mM MgCl_2 , 1 mM PEP, 0.8 mM NADH, pH 8.0, 10 U/mL pyruvate kinase, 16 U/mL lactate dehydrogenase, $1\times$ ATPase buffer.
3. Dilute the ATPase to $2\times$ the final concentration in $1\times$ ATPase buffer along with 2 μM DNA (*see Note 17*). Make an additional mixture without ATPase for a control for background NADH oxidation.
4. Add 60 μL of the $2\times$ reaction cocktail and 60 μL of $2\times$ ATPase in a tube on ice, mixing with a pipette.
5. Add 100 μL of each reaction mixture into the well of a 96-well plate.

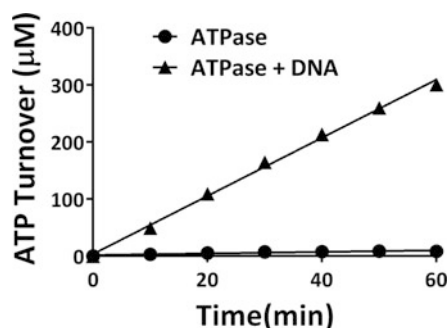


Fig. 3 Representative data of an NADH-coupled ATPase assay. A steady rate of ATP hydrolysis can be quantitated, along with total ATP turnover. Many ATPases involved in DNA metabolism, like the one assayed here, are stimulated by the presence of DNA

6. Insert the 96-well plate into the plate reader. Monitor the absorbance at 340 nm at 37 °C every 60 s for 1 h without shaking (*see Note 18*).
7. Once data collection is complete, export the data to a spreadsheet software (e.g., Microsoft Excel).
8. To calculate the NADH concentration and ultimately the ATP hydrolysis rate for each reaction, an NADH standard curve must be generated (*see Note 19*). Substitute each absorbance read into the $y = mx + b$ equation from the standard curve, where y = absorbance at 340 nm and x = NADH concentration. Subtract the calculated NADH concentrations from the starting NADH concentration. Assuming 1 μM NADH oxidized is equivalent to 1 μM ATP hydrolyzed, these values are the amount of ADP present at each time point. Thus, the data can be presented as ADP generated or ATP hydrolyzed (Fig. 3) (*see Note 20*).

4 Notes

1. TLC buffer should be made fresh for each experiment for optimal resolution.
2. Although this protocol is designed for the measurement of ATPase activity, it can also be used to measure the hydrolysis of other nucleotide triphosphates.
3. In place of a chromatography tank, inexpensive glass tanks can be used instead. A fish tank with a glass lid cut to size works well.
4. Any highly viscous grease (e.g., Vaseline) can be used here.

5. Alternatively, a spectrophotometer can be used to monitor a single reaction.
6. Pencil should always be used instead of ink, which contains pigments that will also separate on the TLC plate. Use light pressure to avoid puncturing the TLC sheet. A blunt, nonmechanical pencil works well.
7. Buffer amount will vary depending on tank size. The buffer should cover the bottom of the tank, but should not rise above 1 cm in depth. Buffer levels higher than the ATP spots will result in diffusion and poor resolution.
8. PCR tubes can also be used. These allow for incubation in a thermocycler but require greater care in ensuring the temperature of the reactions does not rise above ice temperature during reaction assembly.
9. Alternatively, a master mix of components can be assembled, depending on the number of conditions and replicates being assayed. Several master mixes can be assembled for multiple reactions of several conditions (e.g., $\pm$ DNA).
10. This aliquot should be taken as soon as possible after the addition of ATP. Hydrolysis will usually begin to occur even when the samples are on ice.
11. Reactions can be incubated in a gradient of temperatures using an appropriate thermocycler.
12. Time points may need to be taken earlier than 1 min or later than 20 min, depending on the ATPase and reaction conditions.
13. Spots can be aliquoted onto the TLC plate as reactions are stopped or all at once after the reaction has finished. Aliquots are transferred easily if the drop is carefully expelled from the pipette tip, and then gently tapped onto the TLC plate. Spots take approximately 5 min to dry.
14. Depending on the dimensions of your TLC tank, two or more TLC plates can be developed at once in the same tank. The plates should be set up so they are not touching each other, and not overly adhering to the sides of the tank, to ensure proper development.
15. The TLC plate can be placed on top of a few layers of paper towels to facilitate drying.
16. The necessary exposure time is dependent on the activity date of the α [^{32}P]-ATP. Fresh α [^{32}P]-ATP only requires a few hours of exposure, while older ATP may require overnight exposure.
17. Reagents in the $2\times$ reaction cocktail are doubled to provide $1\times$ concentration in the final reaction. Various conditions to assay include ATPase concentration, DNA oligonucleotide sequence

and length, additional cofactors, or other ATPase-specific condition variants. To eliminate any secondary structure the DNA oligonucleotides may form, the DNA can be boiled for 5 minutes, and snap-cooled on ice prior to addition to the cocktail. A reaction should also be run in the absence of DNA, to determine its magnitude of ATPase stimulation, if any.

18. The temperature of the reaction, as well as the frequency and duration of absorption measurements can be adjusted for the ATPase being assayed. Depending on the hydrolysis rate of the ATPase, reactions may need to be equilibrated at room temperature for 3–5 min prior to absorbance measurement on a plate reader. If the reactions go directly from ice to the reaction temperature (e.g., 37 °C), the first few reads will not be linear as the ATPase rate changes along with the temperature of the reactions.
19. To generate an NADH standard curve, measure the absorbance of known concentrations of NADH within a linear range (we use 0.3 to 800 μM). Plot the concentration of NADH on the x-axis, and the absorbance values on the y-axis. Fit the data with a line of best fit. The resultant $y = mx + b$ line equation yields the slope (m) and y-intercept (b) needed to correlate the absorbance values of an ATPase assay with NADH concentration, as in Subheading 3.2, step 8.
20. While ATP is regenerated in this assay, NADH is not. Over the course of the reaction, the absorbance reads will decrease as NADH is oxidized. If all of the NADH is oxidized, the absorbance reads will flat-line, suggesting ATP hydrolysis has ceased. However, these data points should be disregarded as they no longer reflect ATPase activity. For extremely fast ATPases, NADH can be depleted in the time it takes to get the 96-well plate on the plate reader. In this scenario, the concentration of NADH in the reactions should be increased.

Acknowledgments

This work was supported by the College of Arts and Sciences, Indiana University, and Indiana University Collaborative Research Grant fund of the Office of the Vice President for Research, a Collaboration in Translational Research Pilot Grant from the Indiana Clinical and Translational Sciences Institute, and the American Cancer Society [RSG-16-180-01-DMC].

References

1. Heppel LA, Hilmoe RJ (1953) Mechanism of enzymatic hydrolysis of adenosinetriphosphate. *J Biol Chem* 202(1):217–226
2. Ogura T, Wilkinson AJ (2001) AAA(+) superfamily ATPases: common structure-diverse function. *Genes Cells* 6(7):575–597. <https://doi.org/10.1046/j.1365-2443.2001.00447.x>
3. King SM (1995) Chapter 21: Measurement of ATPase activity using [γ -³²P]ATP. In: Dentler W, Witman G (eds) *Methods in cell biology*, vol 47. Academic Press, New York, pp 141–145. [https://doi.org/10.1016/S0091-679X\(08\)60802-X](https://doi.org/10.1016/S0091-679X(08)60802-X)
4. Kagami O, Kamiya R (1995) Chapter 22: Non-radioactive method for ATPase assays. In: Dentler W, Witman G (eds) *Methods in cell biology*, vol 47. Academic Press, New York, pp 147–150. [https://doi.org/10.1016/S0091-679X\(08\)60803-1](https://doi.org/10.1016/S0091-679X(08)60803-1)
5. Rajagopal V, Lorsch JR (2013) Chapter 25: ATP and GTP hydrolysis assays (TLC). In: Lorsch J (ed) *Methods in enzymology*, vol 533. Academic Press, New York, pp 325–334. <https://doi.org/10.1016/B978-0-12-420067-8.00025-8>
6. Kiianitsa K, Solinger JA, Heyer WD (2003) NADH-coupled microplate photometric assay for kinetic studies of ATP-hydrolyzing enzymes with low and high specific activities. *Anal Biochem* 321(2):266–271
7. Andis NM, Sausen CW, Alladin A, Bochman ML (2018) The WYL domain of the PIF1 helicase from the thermophilic bacterium *thermotoga elfii* is an accessory single-stranded DNA binding module. *Biochemistry* 57(7):1108–1118. <https://doi.org/10.1021/acs.biochem.7b01233>
8. Rogers CM, Wang JC, Noguchi H, Imasaki T, Takagi Y, Bochman ML (2017) Yeast Hrq1 shares structural and functional homology with the disease-linked human RecQ4 helicase. *Nucleic Acids Res* 45(9):5217–5230. <https://doi.org/10.1093/nar/gkx151>